DSC640_Kia_Theft_exercise

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

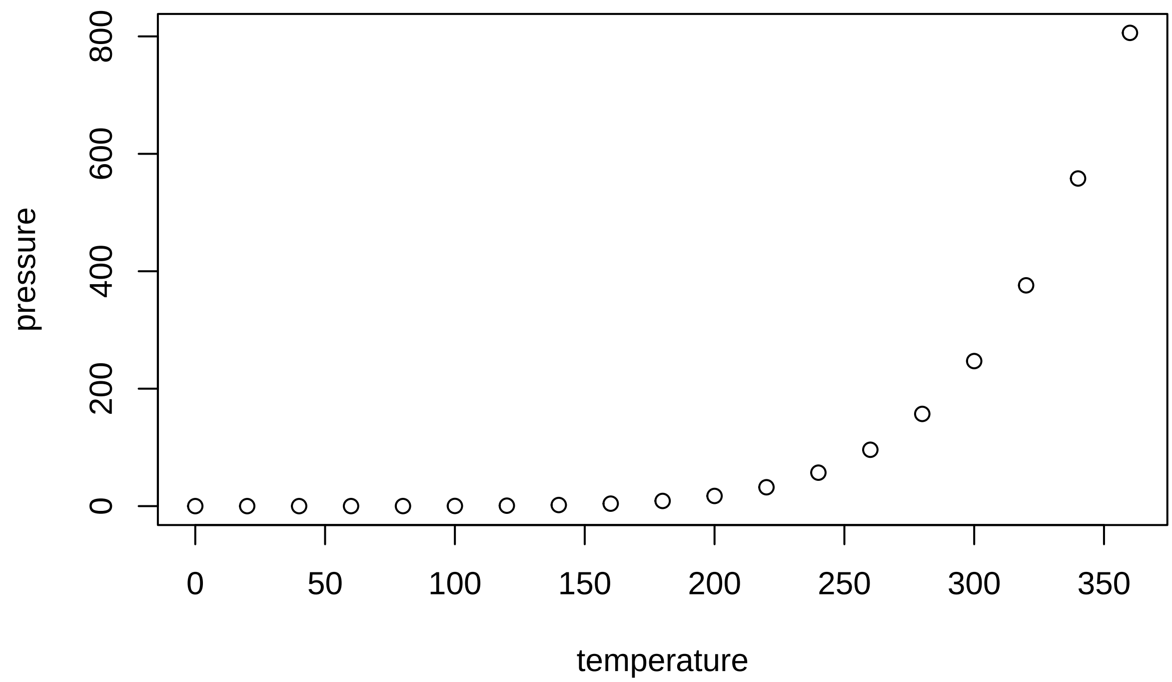
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

library(readxl)

Load the dataset

file_path <- "C:/Users/Teju/Downloads/Motherboard VICE News Kia Hyundai Theft Data.xlsx" df <- read_excel(file_path, sheet = "Data")

View the first few rows

head(df)

Load necessary library

library(readr)

Load the dataset

file_path1 <- "C:/Users/Teju/Downloads/kiaHyundaiThefts.csv" df1 <- read_csv(file_path1)

View the first few rows

head(df1)

Load the dataset

file_path2 <- "C:/Users/Teju/Downloads/carTheftsMap.csv" df2 <- read_csv(file_path2)

View the first few rows

head(df2)

Load the dataset

file_path3 <- "C:/Users/Teju/Downloads/KiaHyundaiMilwaukeeData.csv" df3 <- read_csv(file_path3)

View the first few rows

head(df3)

#1. Pie Chart/Donut Chart – Breakdown of Kia vs. Hyundai Thefts

install.packages("tidyr")

library(ggplot2) library(dplyr) library(readr) library(tidyr)

Load data

kia_hyundai_data <- df1

Summarize thefts by brand

theft_summary <- kia_hyundai_data %>% summarise(Kia = sum(countKiaHyundaiThefts * 0.5), # Assuming equal distribution Hyundai = sum(countKiaHyundaiThefts * 0.5)) %>% pivot_longer(cols = Kia:Hyundai, names_to = "Brand", values_to = "Thefts")

Pie chart

ggplot(theft_summary, aes(x = "", y = Thefts, fill = Brand)) + geom_bar(stat ="identity", width = 1) + coord_polar("y") + theme_void() + labs(title = "Breakdown of Kia vs. Hyundai Thefts") + theme(legend.position = "right")

Donut chart

ggplot(theft_summary, aes(x = ““, y = Thefts, fill = Brand)) + geom_bar(stat =”identity”, width = 0.4) + coord_polar(“y”) + theme_void() + labs(title = “Breakdown of Kia vs. Hyundai Thefts”) + theme(legend.position = “right”)

#2. Stacked Bar Chart – Theft Trends Over Time by City/State

library(ggplot2)

Assuming ‘Year’ is numeric, we create a ‘Date’ column using the first month (January)

kia_hyundai_data$Date < -as.Date(paste(kia_{h}yundai_{d}ata$year, “01”, “01”, sep = “-”))

Proceed with the aggregation and plotting

theft_trend <- kia_hyundai_data %>% group_by(Date, state) %>% summarise(Thefts = sum(countKiaHyundaiThefts, na.rm = TRUE), .groups = “drop”)

Stacked bar chart

ggplot(theft_trend, aes(x = Date, y = Thefts, fill = state)) + geom_bar(stat = “identity”) + theme_minimal() + labs(title = “Kia & Hyundai Theft Trends Over Time by State”, x = “Date”, y = “Number of Thefts”, fill = “State”) + scale_x_date(date_labels = “%Y”, date_breaks = “1 year”) + theme(axis.text.x = element_text(angle = 45, hjust = 1))

3. Tree Map – Most Affected Locations

library(treemap)

Load data

car_thefts <- df2

Aggregate thefts by location

location_summary <- car_thefts %>% group_by(geo_name) %>% summarise(Thefts = sum(countCarThefts2022, na.rm = TRUE))

Tree Map

treemap(location_summary, index = "geo_name", vSize = "Thefts", title = "Most Affected Locations by Car Theft", palette = "Reds")

#4.Bar Chart – Comparison of Kia/Hyundai vs. Other Brands

Aggregate thefts by year

brand_comparison <- kia_hyundai_data %>% group_by(Date) %>% summarise(Kia_Hyundai = sum(countKiaHyundaiThefts, na.rm = TRUE), Other = sum(countOtherThefts, na.rm = TRUE)) %>% pivot_longer(cols = Kia_Hyundai:Other, names_to = "Brand", values_to = "Count")

Bar chart

ggplot(brand_comparison, aes(x = Date, y = Count, fill = Brand)) + geom_col(position = "dodge") + theme_minimal() + labs(title = "Kia/Hyundai Thefts vs Other Brands", x = "Date", y = "Number of Thefts", fill = "Brand")

#Map Visualization – Theft Hotspots

library(ggplot2) library(sf)

Load data

car_thefts_map <- df2

Convert to spatial data

car_thefts_map_sf <- st_as_sf(car_thefts_map, coords = c("longitude", "latitude"), crs = 4326)

Map visualization

ggplot() + geom_sf(data = car_thefts_map_sf, aes(size = countCarThefts2022), color = "red", alpha = 0.6) + theme_minimal() + labs(title = "Geographic Distribution of Car Thefts", size = "Number of Thefts")